1 Linear Algebra (Theory & Algorithms)

1.1 Vector Spaces & Subspaces

Vector Space axioms: Closure under addition and scalar multiplication, associativity, commutativity, identity, inverses, distributive laws. **Subspace Test:** A subset W of a vector space V is a subspace if: 1. The zero vector $\mathbf{0} \in W$. 2. For all $u, v \in W$, $u + v \in W$ (Closed under addition). 3. For all $u \in W, c \in \mathbb{R}$, $cu \in W$ (Closed under scalar mult).

Basis: A set of vectors $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis for V if:

- They correspond to linearly independent vectors.
- Their span covers V ($\text{Span}(\mathcal{B}) = V$).

Dimension: The number of vectors in any basis of V.

1.2 Linear Maps, Image, & Kernel

Let $T : V \to W$ be a linear transformation ($T(x) = Ax$).

- **Kernel (Null Space):** $\text{Ker}(T) = \{x \in V \mid T(x) = 0\}$.
- **Image (Range/Col Space):** $\text{Im}(T) = \{w \in W \mid \exists x \in V, T(x) = w\}$.
- **Rank:** $\dim(\text{Im}(T))$.
- **Rank-Nullity Theorem:**

$$\dim(V) = \dim(\text{Ker}(T)) + \dim(\text{Im}(T))$$

$$(\#\text{ Cols}) = (\#\text{ Free Vars}) + (\#\text{ Pivot Cols})$$

1.3 Orthogonal Diagonalization (Algorithm)

Required for Symmetric Matrices ($A = A^T$). **Theorem (Spectral Theorem):** If A is symmetric, then A is orthogonally diagonalizable ($A = PDP^T$ with P orthogonal). **Step-by-Step Process:**

1. **Find Eigenvalues:** Solve $\det(A - \lambda I) = 0$.
2. **Find Eigenspaces:** For each λ, find the basis for $\text{Nul}(A - \lambda I)$.
3. **Gram-Schmidt Process:** If an eigenvalue has multiplicity $k > 1$, the basis vectors $v_1, \dots, v_k$ might not be orthogonal. Orthogonalize them:

$$u_1 = v_1$$
$$u_2 = v_2 - \text{proj}_{u_1} v_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1$$

4. **Normalize:** Convert all u_i to unit vectors: $e_i = \frac{u_i}{||u_i||}$.
5. **Construct P and D:** $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ $P = [e_1 \dots e_n]$ (Columns are orthonormal eigenvectors). Check: $P^T P = I$.

1.4 Singular Value Decomposition (SVD)

Let A be an $m \times n$ matrix with rank r. Then there exists an $m \times n$ factorization:

$$A = U \Sigma V^T$$

- U: $m \times m$ orthogonal matrix (Left Singular Vectors).
- Σ: $m \times n$ diagonal matrix with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$.
- V: $n \times n$ orthogonal matrix (Right Singular Vectors).

Simple Example: Find SVD of $A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$.

1. **Compute $A^T A$:** $\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}$.
2. **Eigenvalues of $A^T A$:** $\det(A^T A - \lambda I) = \lambda^2 - 10\lambda$. $\lambda_1 = 10, \lambda_2 = 0$.
3. **Singular Values:** $\sigma_1 = \sqrt{10}, \sigma_2 = 0$. $\Sigma = \begin{pmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{pmatrix}$.
4. **V (Eigenvectors of $A^T A$):** $\lambda = 10$: $\text{Nul} \begin{pmatrix} -5 & 5 \\ 5 & -5 \end{pmatrix} \implies v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. $\lambda = 0$: $\text{Nul} \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix} \implies v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
5. **U (Vectors in $\mathbb{R}^m$):** $u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{20}} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. u_2 must be unit, orthogonal to u_1. $u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

2 Probability (Foundations & Theorems)

2.1 Limit Theorems (Exact Formulation)

Theorem 2.126 (Weak Law of Large Numbers) *Let $X_1, X_2, \dots, X_n$ be a random sample of size n all with the same finite mean μ and variance σ^2. If $\overline{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$, then for any constant $c > 0$,*

$$\boxed{\lim_{n \to \infty} P(|\overline{X} - \mu| \geq c) = 0.}$$

Theorem 2.122 (Central Limit Theorem) *Let $X_1, X_2, \dots$ be a sequence of independent rvs all having the same distributions, $E(X_1) = \mu$, and $Var(X_1) = \sigma^2$. Then for any $a, b \in \mathbb{R}$,*

$$\boxed{\lim_{n \to \infty} P\left(a \leq \frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq b\right) = P(a \leq Z \leq b)}$$

where $Z \sim N(0, 1)$.

2.2 Axioms & Set Theory

De Morgan's Laws: $(A \cup B)^c = A^c \cap B^c$. $(A \cap B)^c = A^c \cup B^c$. **Inclusion-Exclusion:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. **Partition Rule:** If $B_1, \dots, B_n$ partition the sample space S:

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A|B_i) P(B_i)$$

2.3 Conditional Probability & Bayes

Bayes' Theorem:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Used for updating beliefs (e.g., Mock Ex 5 "Coin Problem"). *Independence:* Events A, B are independent iff $P(A \cap B) = P(A)P(B)$.

2.4 Moment Generating Functions (MGF)

Definition: $M_X(t) = E[e^{tX}]$. **Properties:**

- $M_X(0) = 1$.
- **Moments:** The n-th derivative at $t = 0$ gives the n-th moment:

$$M_X^{(n)}(0) = E[X^n]$$

- **Sum of Indep. RVs:** If $Y = X_1 + \dots + X_n$, then $M_Y(t) = \prod M_{X_i}(t)$.
- **Uniqueness:** If two RVs have the same MGF, they have the same distribution.

2.5 Fundamental Functions (PMF, PDF, CDF)

PMF (Discrete): Probability Mass Function, $p_X(x) = P(X = x)$.

- $\sum_{\text{all } x} p_X(x) = 1$ and $0 \leq p_X(x) \leq 1$.

PDF (Continuous): Probability Density Function, $f_X(x)$.

- $P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$.
- $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$ and $f_X(x) \geq 0$.
- *Note:* $f_X(x)$ is NOT a probability (can be > 1).

CDF (Cumulative): Cumulative Distribution Function, $F_X(x) = P(X \leq x)$.

- Properties: Non-decreasing, $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$.
- Relation to PDF: $f_X(x) = \frac{d}{dx} F_X(x)$ (by FTC).

3 Distributions (Discrete & Continuous)

3.1 Discrete Random Variables

Bernoulli(p):

- PMF: $P(X = 1) = p, P(X = 0) = 1 - p$.
- Mean: p. Var: $p(1 - p)$.

Binomial(n, p): (Sum of n i.i.d. Bernoullis)

- PMF: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
- Mean: np. Var: $np(1 - p)$.

Geometric(p): (Trials until first success)

- PMF: $P(X = k) = (1 - p)^{k-1} p$.
- Mean: $1/p$. Var: $(1 - p)/p^2$.

Poisson(λ): (Rate of events)

- PMF: $P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$ for $k = 0, 1, \ldots$
- Mean: λ. Var: λ.
- MGF: $e^{\lambda(e^t - 1)}$.

3.2 Continuous Random Variables

Uniform(a, b):

- PDF: $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$.
- Mean: $\frac{a+b}{2}$. Var: $\frac{(b-a)^2}{12}$.

Exponential(λ): (Waiting time)

- PDF: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
- CDF: $F(x) = 1 - e^{-\lambda x}$.
- Mean: $1/\lambda$. Var: $1/\lambda^2$.
- **Memoryless Property:** $P(X > s + t | X > s) = P(X > t)$.

Normal(μ, σ^2):

- PDF: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.
- Standard Normal $Z \sim N(0, 1)$. Transformation: $Z = \frac{X-\mu}{\sigma}$.
- MGF: $e^{\mu t + \frac{1}{2}\sigma^2 t^2}$.

3.3 Joint Distributions & Covariance

Joint PDF: $P((X, Y) \in A) = \iint_A f(x, y) dx dy$. **Marginal PDF:** $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$. **Covariance:**

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y]$$

Correlation: $\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$. **Variance of Sum:**

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

If X, Y are independent, $Cov(X, Y) = 0$ and $\rho = 0$.